

Sentinel-2 L2A NDVI time series
(cell-level, Kharif window 2019-2024)

Stage A — Dekadal NDVI compositing
(10-day max-NDVI; ESA WorldCover crop mask;
Sentinel-2 SCL cloud screening)

Stage B — Double-logistic phenometric extraction
(Whittaker smoother → 6-parameter fit;
SOS / POS / EOS retrieval)

QC FRAMEWORK (the primary contribution of this paper)

GATE A

Mode-share threshold

Panel-wide mode-share
(fraction at single DOY)
 ≤ 0.20 for SOS, POS, EOS

v1.0.2 (raw):
SOS 0.203 • POS 0.656 • EOS 0.727
v2.0 (refit):
SOS 0.125 • POS 0.083 • EOS 0.083

GATE B

Biological plausibility

Median DOY within Kharif rice
agronomic windows:
SOS $\in [155, 240]$
POS $\in [240, 320]$
EOS $\in [280, 380]$

v2.0 observed ranges:
SOS 168-241 • POS 256-304 • EOS 320-374

GATE C

Fit-quality screening

Per-pixel RMSR ≤ 0.15 NDVI units
Cell excluded if $> 30\%$ of pixels
flagged as poor-fit

v2.0 panel-avg fit-fail rate:
0.61 (no cell $> 30\%$)

All three gates passed?

yes

PASS

→ District-year cell enters
DiD estimation panel
(n = 48; Models 1 and 2)

no

FAIL

→ Cell flagged and excluded
from DiD panel
(reason code logged)

EMPIRICAL VALIDATION — Odisha panel, 2019-2024, 8 districts × 6 years

EOS mode-share
0.727 → 0.083

POS mode-share
0.656 → 0.083

Unique EOS DOY values
2 → 38